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The dual polarization of an abelian variety

By

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Abstract. Suppose (X, H) is a polarized abelian variety. There are several possibilities to define a polarization on the dual abelian variety $\hat{X}$ induced by H: 1) use the isogeny $X \longrightarrow \hat{X}$ defined by H, 2) take the image of H under the Lefschetz operator associated with H and apply Poincaré duality, 3) consider the image in $\hat{X}$ of a theta divisor for H on X, 4) use Mukai's Fourier transform. In the paper these four dual polarizations are introduced and it is shown that they essentially coincide.

Let X denote a complex abelian variety of dimension g. A *polarization* on X is by definition the first Chern class $H = c_1(L) \in H^2(X, \mathbb{Z})$ of an ample line bundle L on X. By abuse of notation we also call the line bundle L itself a polarization. The number $d = h^0(L)$ is called the *degree* of the polarization L. Let $\hat{X} := \mathrm{Pic}^0(X)$ denote the dual abelian variety. There are several possibilities to associate to the polarization H on X a polarization on the dual abelian variety:

(1) The homomorphism $\phi_L : X \to \hat{X},\ x \mapsto t_x^* L \otimes L^{-1}$ is an isogeny of degree d^2. There is a unique isogeny $\psi : \hat{X} \to X$ such that $\psi \circ \phi_L = d \cdot \mathrm{id}_X$. It is easy to see that there is a unique polarization $\hat{H}_1 = c_1(\hat{L}_1)$ on the dual abelian variety $\hat{X}$ such that $\psi = \phi_{\hat{L}_1}$.

(2) Let $\alpha : H^2(X, \mathbb{Z}) \to H^2(\hat{X}, \mathbb{Z})$ denote the composed map $H^2(X, \mathbb{Z}) \xrightarrow{\mathbb{L}^{g-2}} H^{2g-2}(X, \mathbb{Z}) \xrightarrow{P^{-1}} H_2(X, \mathbb{Z}) \xrightarrow{\delta_2} H^2(\hat{X}, \mathbb{Z})$, where $\mathbb{L}$ denotes the Lefschetz operator, $P : H_2(X, \mathbb{Z}) \to H^{2g-2}(X, \mathbb{Z})$ the Poincaré duality and δ_2 the canonical duality. We will see that $\hat{H}_2 := \alpha(H)$ is a polarization on $\hat{X}$.

(3) Let D denote a divisor in the linear system $|L|$. The divisor $\phi_{L*}(D)$ defines a polarization $\hat{H}_3$ on $\hat{X}$, which only depends on H and not on the special choice of D and L.

(4) If $\mathscr{F}$ denotes Mukai's Fourier transform, then $\mathscr{F}(L)$ is a vector bundle of rank d on $\hat{X}$. We will see that the line bundle $\hat{L}_4 := \det(F(L))^{-1}$ defines a polarization $\hat{H}_4 := c_1(\hat{L}_4)$ on $\hat{X}$.

The main result of the paper is the following theorem, which shows that the polarizations $\hat{H}_i,\ i = 1, \ldots, 4$ essentially coincide.

Theorem. $\hat{H}_1 = \dfrac{1}{(g-1)!}\,\hat{H}_2 = \dfrac{1}{d}\hat{H}_3 = \hat{H}_4$ *for any polarization* H *on* X.

Hence it makes sense to call the polarization $\hat{H} := \hat{H}_1 = \hat{H}_4$ *the dual polarization of* H on $\hat{X}$.

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The main idea for the proof of the theorem is to express the involved isomorphisms and dualities in terms of a fixed symplectic basis for L and then to compute the cohomology classes $\hat{H}_i$ explicitly. In the first section we recall the basic isomorphisms and dualities and in the second section we prove the above theorem.

1. Preliminaries. In this section we recall some identifications and dualities between homology and cohomology groups, needed subsequently. For details we refer to [3].

(a) The symplectic basis. Let (X, L) denote a *polarized complex abelian variety* of dimension g, that is X is a complex torus $X = V/\Lambda$, where V is a complex vector space of dimension g and Λ a lattice in V, and L is an ample line bundle defining a polarization $H = c_1(L) \in H^2(X, \mathbb{Z})$ on X. H may be considered as a positive definite hermitian form on the vector space V, whose imaginary part $E = \operatorname{Im} H$ is integer valued on the lattice Λ. If $d_1, \ldots, d_g$, $0 < d_i | d_{i+1}$ denote the elementary divisors of the alterning form E on the lattice Λ, then the line bundle L and the polarization H are called *of type* $(d_1, \ldots, d_g)$. A basis $\lambda_1, \ldots, \lambda_{2g}$ of Λ is called *symplectic basis* for L or H, if

(1) $$E(\lambda_i, \lambda_j) = d_i\, \delta_{g+i,j}$$

for all $i \leqq j$. There is a canonical identification

(2) $$H_1(X, \mathbb{Z}) = \Lambda.$$

Fix once and for all a symplectic basis $\lambda_1, \ldots, \lambda_{2g}$ of $\Lambda = H_1(X, \mathbb{Z})$ and denote by $x_1, \ldots, x_{2g}$ the corresponding real coordinate functions on V. By (1) we have for all $i = 1, \ldots, g$

(3) $$x_i(\cdot) = -\frac{1}{d_i} E(\lambda_{g+i}, \cdot) \ \text{ and } \ x_{g+i}(\cdot) = \frac{1}{d_i} E(\lambda_i, \cdot)$$

as functions on the vector space V.

(b) The dual abelian variety. Let $\bar{V}^* := \operatorname{Hom}_{\overline{\mathbb{C}}}(V, \mathbb{C})$ denote the vector space of anti-linear forms on V and

$$\hat{\Lambda} := \{l \in \bar{V}^* | \operatorname{Im} l(\Lambda) \subseteqq \mathbb{Z}\}$$

the dual lattice in $\bar{V}^*$. The complex torus $\hat{X} = \bar{V}^*/\hat{\Lambda}$ is an abelian variety, called the *dual abelian variety*. Let $x_1, \ldots, x_{2g} \in \operatorname{Hom}_{\mathbb{R}}(V, \mathbb{R})$ again denote the real coordinate functions given by the symplectic basis. Using the canonical isomorphism

$$\operatorname{Hom}_{\mathbb{R}}(V, \mathbb{R}) \xrightarrow{\sim} \operatorname{Hom}_{\overline{\mathbb{C}}}(V, \mathbb{C}) = \bar{V}^*\ , \ y(\cdot) \mapsto -y(i\,\cdot) + iy(\cdot)$$

we see that $\varepsilon_1, \ldots, \varepsilon_{2g}$, defined by

(4) $$\varepsilon_j(\cdot) = -x_j(i\,\cdot) + ix_j(\cdot)$$

form a basis of the lattice $\hat{\Lambda}$. According to (3) we have for $j = 1, \ldots, g$:

(5) $$\varepsilon_j(\cdot) = -\frac{1}{d_j} H(\lambda_{g+j}, \cdot) \quad \text{and} \quad \varepsilon_{g+j} = \frac{1}{d_j} H(\lambda_j, \cdot),$$

since $H(\cdot,\cdot) = E(i\cdot,\cdot) + iE(\cdot,\cdot)$ and $E(i\cdot, i\cdot) = E(\cdot,\cdot)$. Let $a_1, \ldots, a_{2g} \in \operatorname{Hom}_{\mathbb{R}}(\bar{V}^*, \mathbb{R})$ denote the corresponding real coordinate functions. Identifying $\Lambda = \hat{\hat{\Lambda}}$ and $V = \overline{(\overline{V}^*)}^*$, we clearly have

$$\lambda_j(\cdot) = -a_j(i\cdot) + ia_j(\cdot)$$

The map $\phi_H : V \to \bar{V}^*,\ v \mapsto H(v, \cdot)$ relates the coordinate functions x_j and a_k as follows:

Lemma 1.1. *$\phi_H^* a_k = -d_k x_{g+k}$ and $\phi_H^* a_{g+k} = d_k x_k$, for any $k = 1, \ldots, g$.*

Proof. For $l = 1, \ldots, g$ we have using (5)

$$\phi_H^* a_k(\lambda_{g+l}) = a_k(\phi_H(\lambda_{g+l})) = -d_l a_k(\varepsilon_l) = -d_l \delta_{kl}.$$

The same computation yields $\phi_H^* a_k(\lambda_l) = 0$ for all $l = 1, \ldots, g$. Since the coordinate function $x_\mu : V \to \mathbb{R}$ is characterized by $x_\mu(\lambda_\nu) = \delta_{\mu\nu}$, this implies $\phi_H^* a_k = -d_k x_{g+k}$. The proof of the second equation is the same. $\square$

(c) Pontrjagin product and cup product. Let us denote by $\times : H_p(X, \mathbb{Z}) \times H_p(X, \mathbb{Z}) \to H_{p+q}(X \times X, \mathbb{Z})$ the exterior homology product. The addition map $\mu : X \times X \to X$ induces a homomorphism $\mu_* : H_{p+q}(X \times X, \mathbb{Z}) \to H_{p+q}(X, \mathbb{Z})$. The *Pontrjagin product* on X is defined to be the composition

$$\star : H_p(X, \mathbb{Z}) \times H_q(X, \mathbb{Z}) \xrightarrow{\times} H_{p+q}(X \times X, \mathbb{Z}) \xrightarrow{\mu_*} H_{p+q}(X, \mathbb{Z}).$$

It induces an isomorphism for every $n \geqq 1$

(6) $$\star_{i=1}^n : H_1(X, \mathbb{Z}) \xrightarrow{\sim} H_n(X, \mathbb{Z}).$$

Note that the Pontrjagin product commutes with f_* for every homomorphism of abelian varieties $f : X \to Y$:

(7) $$f_*(\alpha) \star f_*(\beta) = f_*(\alpha \star \beta)$$

for every $\alpha \in H_p(X, \mathbb{Z})$, $\beta \in H_q(X, \mathbb{Z})$. This follows immediately from the definition, since the exterior product and μ_* are compatible with f_*.

Similar to the identification (2) there is an identification

(8) $$H^1(X, \mathbb{Z}) = \mathrm{Hom}(\Lambda, \mathbb{Z})$$

and the *cup product* $\wedge : H^p(X, \mathbb{Z}) \times H^q(X, \mathbb{Z}) \to H^{p+q}(X, \mathbb{Z})$ induces an isomorphism for every $n \geqq 1$

(9) $$\wedge^n H^1(X, \mathbb{Z}) \simeq H^n(X, \mathbb{Z}).$$

The Pontrjagin product is dual to the cup product. The precise meaning of this statement will be given in the next subsection (see diagram (10)).

(d) The duality isomorphism *D*. The differentials $dx_1, \ldots, dx_{2g}$ of the coordinate functions $x_1, \ldots, x_{2g}$ form a basis of $H^1(X, \mathbb{Z})$, dual to $\lambda_1, \ldots, \lambda_{2g}$, that is

$$\int_{\lambda_i} dx_j = \delta_{ij}.$$

Of course, the coordinate functions x_i being linear, we can identify x_i with dx_i and consider

$x_1, \ldots, x_{2g}$ as the basis of $H^1(X, \mathbb{Z})$ by (8), but for obvious reasons we prefer the notation dx_i here. For any ordered multi-index $I = (1 \leqq i_1 < \ldots < i_p \leqq 2g)$ we write

$$\lambda_I := \lambda_{i_1} \star \ldots \star \lambda_{i_p} \quad \text{and} \quad dx_I := dx_{i_1} \wedge \ldots \wedge dx_{i_p}.$$

Then, according to [3], Lemma 4.10.1, the sets $\{\lambda_I | \sharp I = p\}$ and $\{dx_I | \sharp I = p\}$ form bases of $H_p(X, \mathbb{Z})$ and $H^p(X, \mathbb{Z})$ respectively, dual to each other with respect to the natural isomorphism

$$H_p(X, \mathbb{Z}) \to H^p(X, \mathbb{Z})^*, \ \sigma \mapsto \int_\sigma .$$

Define the duality isomorphism D by

$$D : H_p(X, \mathbb{Z}) \to H^p(X, \mathbb{Z}), \quad \lambda_I \mapsto dx_I.$$

Note that D depends not only on the polarization but also on the choice of the symplectic basis. It is now easy to see that the following diagram commutes:

$$\begin{array}{ccc} H_p(X,\mathbb{Z}) \times H_q(X,\mathbb{Z}) & \xrightarrow{\star} & H_{p+q}(X,\mathbb{Z}) \\ {\scriptstyle D\times D}\downarrow & & \downarrow{\scriptstyle D} \\ H^p(X,\mathbb{Z}) \times H^q(X,\mathbb{Z}) & \xrightarrow{\wedge} & H^{p+q}(X,\mathbb{Z}) \end{array} . \tag{10}$$

(e) Poincaré duality. The Poincaré duality isomorphism

$$P : H_p(X, \mathbb{Z}) \to H^{2g-p}(X, \mathbb{Z})$$

is expressed in terms of the bases $\{\lambda_I\}$ of $H_p(X, \mathbb{Z})$ and $\{dx_j\}$ of $H^{2g-p}(X, \mathbb{Z})$ as follows (see [3], Section 4.10): For any multi-index $I = (i_1 < \ldots < i_p)$ of $\{1, \ldots, 2g\}$ denote by $I^0 := (i_1^0 < \ldots < i_{2g-p}^0)$ the ordered multi-index, for which as a set $I \cup I^0 = \{1, \ldots, 2g\}$, and define the sign $\varepsilon(I)$ of I by

$$\varepsilon(I) dx_I \wedge dx_{I^0} = dx_1 \wedge dx_{g+1} \wedge \ldots \wedge dx_g \wedge dx_{2g}.$$

Then $P : H_p(X, \mathbb{Z}) \to H^{2g-2}(X, \mathbb{Z})$ is given by

$$P(\lambda_I) = (-1)^{g+p} \varepsilon(I) dx_{I^0} \tag{11}$$

for every ordered multi-index I of length p. By the original definition (see [3]) the isomorphism P does not depend on the choice of the symplectic basis.

(f) The Fourier transform φ_p. The canonical pairing

$$\Lambda \times \hat{\Lambda} \to \mathbb{Z}, \ (\lambda, l) \mapsto \operatorname{Im} l(\lambda)$$

is nondegenerate and hence yields an isomorphism

$$\delta_1 : H_1(X, \mathbb{Z}) \to H^1(\hat{X}, \mathbb{Z}).$$

The differentials $da_1, \ldots, da_{2g}$ of the coordinate functions a_j of subsection (b) form a basis of $H^1(\hat{X}, \mathbb{Z})$ and by construction δ_1 is given by

$$\delta_1(\lambda_j) = da_j$$

for $j = 1, \ldots, 2g$. Using equations (6) and (9) δ_1 induces an isomorphism

$$\delta_p : H_p(X, \mathbb{Z}) \to H^p(\hat{X}, \mathbb{Z})$$

for any p in a canonical way. For every multi-index $I = (i_1 < \ldots < i_p)$ we have $\delta_p(\lambda_I) = da_I$. Define the isomorphism

$$\varphi_p : H^p(X, \mathbb{Z}) \to H^{2g-p}(\hat{X}, \mathbb{Z})$$

to be the composed map

$$\varphi_p : H^p(X, \mathbb{Z}) \xrightarrow{P^{-1}} H_{2g-p}(X, \mathbb{Z}) \xrightarrow{\delta_{2g-p}} H^{2g-p}(\hat{X}, \mathbb{Z}).$$

Using (9) and (11) and the fact that $\varepsilon(I^0) = (-1)^p \varepsilon(I)$ we have

$$\varphi_p(dx_I) = (-1)^g \varepsilon(I) da_{I^0}. \tag{12}$$

In [1] Beauville introduced a cohomological version $\mathscr{F}_p$ of Mukai's Fourier transform in the following way: Let $\mathscr{P}$ denote the Poincaré bundle on $X \times \hat{X}$ and p_1 and p_2 the projections of $X \times \hat{X}$ onto its factors. Then $\mathscr{F}_p : H^p(X, \mathbb{Z}) \to H^{2g-p}(\hat{X}, \mathbb{Z})$ is defined by

$$\mathscr{F}_p(z) = p_{2*}(e^{c_1(\mathscr{P})} \wedge p_1^*(z))$$

for all $z \in H^p(X, \mathbb{Z})$.

Proposition 1.2. $\varphi_p = (-1)^{\frac{1}{2}p(p+1)} \mathscr{F}_p$.

Proof. First one shows that

$$\mathscr{F}_p(dx_I) = (-1)^{\frac{1}{2}(2g-p)(2g-p-1)} \varepsilon(I) da_{I^0} \tag{13}$$

for every multi-index $I = (i_1 < \ldots < i_p)$. We omit the proof, since it is the same as the proof of the analogous equation in the proof of Proposition 1 of [1]. Note however that Beauville works with a basis of Λ which provides the natural orientation of the complex vector space V, instead of a symplectic basis. In particular his isomorphism $H^{2g}(X, \mathbb{Z}) \xrightarrow{\sim} \mathbb{Z}$ differs by a sign from our isomorphism. One could also apply [3], Lemma 3.6.5 to deduce (13) directly from [1]. The proof of Proposition 1.2 is completed by comparing (12) and (13). □

2. Proof of the theorem. Let L be a line bundle defining a polarization H of type $(d_1, \ldots, d_g)$ on $X = V/\Lambda$. Fix a symplectic basis $\lambda_1, \ldots, \lambda_{2g}$ of Λ for L and denote by $x_1, \ldots, x_{2g}$ the corresponding real coordinate functions on V. On the one hand the first Chern class of L may be considered as a hermitian form on the vector space V, on the other hand it may be considered as an element of $H^2(X, \mathbb{Z})$ or, using equation (8) and (9), as an alternating form on the lattice Λ. In order to distinguish the different points of view in the notation, we write H in the first case and $c_1(L)$ in the latter case. In these terms we have with respect to the basis $\{dx_i \wedge dx_j | 1 \leqq i < j \leqq 2g\}$ of $H^2(X, \mathbb{Z})$

$$c_1(L) = -\sum_{\nu=1}^{g} d_\nu dx_\nu \wedge dx_{g+\nu} \tag{14}$$

(see [3], Lemma 3.6.4). We use this to compute the alternating form of the polarizations $\hat{H}_1, \ldots, \hat{H}_4$ of the introduction.

(a) The polarization $\hat{H}_1$. The map $\phi_L : X \to \hat{X} = \mathrm{Pic}^0(X)$, $x \mapsto t_x^* L \otimes L^{-1}$ is an isogeny of degree $d := d_1 \dots d_g$. Since ker ϕ_L consists of d-division points of X, there is a unique isogeny $\psi : \hat{X} \to X$ such that $\psi \circ \phi_L = d \cdot \mathrm{id}_X$ and $\phi_L \circ \psi = d \cdot \mathrm{id}_{\hat{X}}$.

Proposition 2.1. *There is a line bundle $\hat{L}_1$ defining a polarization $\hat{H}_1$ on $\hat{X}$ such that $\psi = \phi_{\hat{L}_1}$.*

Proof. Let $F : \bar{V}^* \to V$ denote the analytic representation of ψ. According to [3] Theorem 2.5.5 it suffices to show that the associated form (which we also denote by F) $F : \bar{V}^* \times \bar{V}^* \to \mathbb{C}$, $(l_1, l_2) \mapsto l_2(F(l_1))$ is hermitian.

Since $\phi_H : V \to \bar{V}^*$, $v \mapsto H(v, \cdot)$ is the analytic representation of $\phi_L : X \to \hat{X}$, we have $F \circ \phi_H = d \cdot \mathrm{id}_V$ and thus

$$F = d\phi_H^{-1} : \bar{V}^* \to V.$$

For any $\nu = 1, \dots, g$

$$\begin{aligned} \phi_H(\lambda_{g+\nu})(\cdot) &= H(\lambda_{g+\nu}, \cdot) \\ &= -E(\lambda_{g+\nu}, i\cdot) + iE(\lambda_{g+\nu}, \cdot) \\ &= d_\nu x_\nu(i\cdot) - d_\nu i x_\nu(\cdot) \qquad \text{(by (3))} \\ &= -d_\nu \varepsilon_\nu(\cdot) \qquad \text{(by (4))}. \end{aligned}$$

Hence we have for the form $F : \bar{V}^* \times \bar{V}^* \to \mathbb{C}$ and all $1 \leqq \nu, \mu \leqq g$

$$\begin{aligned} F(\varepsilon_\nu, \varepsilon_\mu) &= F(\varepsilon_\nu)(\varepsilon_\mu) \\ &= d\phi_H^{-1}(\varepsilon_\nu)(\varepsilon_\mu) \\ &= -\frac{d}{d_\nu}\lambda_{g+\nu}(\varepsilon_\mu) \qquad \text{where} \quad \lambda_\nu \in \overline{(\overline{V}^*)}^* = V \\ &= -\frac{d}{d_\nu}\varepsilon_\mu(\lambda_{g+\nu}) \\ &= \frac{d}{d_\nu d_\mu} H(\lambda_{g+\mu}, \lambda_{g+\nu}) \qquad \text{(by (5))}. \end{aligned}$$

This implies that the matrix of the map F with respect to the $\mathbb{C}$-basis $\varepsilon_1, \dots, \varepsilon_g$ of $\bar{V}^*$ is hermitian, which completes the proof of Proposition 2.1. $\square$

Proposition 2.2. *$\hat{H}_1$ is the unique polarization of $\hat{X}$ such that*

$$\phi_L^* \hat{H}_1 = dH. \tag{15}$$

Proof. The pullback map $\phi_L^* : \mathrm{NS}(\hat{X}) \to \mathrm{NS}(X)$ is injective, since $\phi_L : X \to \hat{X}$ is an isogeny. Hence it suffices to verify (15). But $\phi_{\phi_L^* \hat{L}_1}$ fits into the following commutative diagram (see [3], Corollary 2.4.6)

$$\begin{array}{ccc} \hat{X} & \xrightarrow{\phi_{\hat{L}_1}} & X = \hat{\hat{X}} \\ {\scriptstyle \phi_L}\uparrow & & \downarrow{\scriptstyle \phi_L} \\ X & \xrightarrow{\phi_{\phi_L^* \hat{L}_1}} & \hat{X} \end{array} .$$

Hence $\phi_{\phi_L^* \hat{L}_1} = \phi_L \, \phi_{\hat{L}_1} \phi_L = d\phi_L$. By [3], Proposition 2.5.3 this is equivalent to $\phi_L^* \hat{H}_1 = c_1(\phi_L^* \hat{L}_1) = c_1(L^d) = dH$. □

Now we are in a position to compute $c_1(\hat{L}_1) \in H^2(\hat{X}, \mathbb{Z})$ in terms of the basis $\{da_k \wedge da_l | 1 \leqq k < l \leqq 2g\}$ of $H^2(\hat{X}, \mathbb{Z})$ of Section 1.

Proposition 2.3. $c_1(\hat{L}_1) = -\sum_{\nu=1}^{g} \frac{d}{d_\nu} da_\nu \wedge da_{g+\nu}$.

Proof. According to Proposition 2.2 and (14) and since ϕ_H is the analytic representation of ϕ_L we have to show

$$\phi_H^* \left(-\sum_{\nu=1}^{g} \frac{d}{d_\nu} da_\nu \wedge da_{g+\nu} \right) = -d \sum_{\nu=1}^{g} d_\nu dx_\nu \wedge dx_{g+\nu}.$$

But this follows immediately from Lemma 1.1. Note that x_ν and a_ν are linear functions and thus we may identify them with dx_ν and da_ν. □

(b) The polarization $\hat{H}_2$. The Lefschetz operator $\mathbb{L}$ associated to the polarization defined by the line bundle L on X is given by

$$\mathbb{L} : H^p(X, \mathbb{Z}) \to H^{p+2}(X, \mathbb{Z}), \quad \varphi \mapsto c_1(L) \wedge \varphi.$$

Consider the composed map

$$\alpha : H^2(X, \mathbb{Z}) \xrightarrow{\mathbb{L}^{g-2}} H^{2g-2}(X, \mathbb{Z}) \xrightarrow{\varphi_{2g-2}} H^2(\hat{X}, \mathbb{Z})$$

where φ_{2g-2} denotes the Fourier transform of Section 1 (f).

Proposition 2.4. (i) *$\alpha(c_1(L))$ is the class of a polarization $\hat{L}_2$ on $\hat{X}$.*

(ii) $c_1(\hat{L}_2) = -(g-1)! \sum_{\nu=1}^{g} \frac{d}{d_\nu} da_\nu \wedge da_{g+\nu}$.

Proof.

$$\begin{aligned}
\alpha(c_1(L)) &= \varphi_{2g-2}(c_1(L)^{\wedge(g-1)}) \\
&= \varphi_{2g-2}\left(\left(-\sum_{\nu=1}^{g} d_\nu dx_\nu \wedge dx_{g+\nu} \right)^{\wedge(g-1)} \right) \\
&= \varphi_{2g-2}\Bigg((-1)^{g-1}(g-1)! \sum_{\nu=1}^{g} \frac{d}{d_\nu} dx_1 \wedge dx_{g+1} \wedge \ldots \\
&\qquad\qquad \wedge d\check{x}_\nu \wedge d\check{x}_{g+\nu} \wedge \ldots \wedge dx_g \wedge dx_{2g} \Bigg) \\
&= (-1)^{2g-1}(g-1)! \sum_{\nu=1}^{g} \frac{d}{d_\nu} da_\nu \wedge da_{g+\nu}
\end{aligned}$$

according to (12). Proposition 2.3 implies

$$\alpha(c_1(L)) = (g-1)! c_1(\hat{L}_1) = c_1(\hat{L}_1^{\otimes(g-1)!})$$

This implies (i) and (ii). □

(c) The polarization $\hat{H}_3$. Let D denote a divisor in the linear system $|L|$ and let $\hat{H}_3$ denote the class of the push forward $\phi_{L_*}(D)$ in $H^2(\hat{X}, \mathbb{Z})$.

Proposition 2.5. (i) *$\hat{H}_3$ is a polarization on $\hat{X}$, that is there is an ample line bundle $\hat{L}_3$ on $\hat{X}$ such that $\hat{H}_3 = c_1(\hat{L}_3)$.*

(ii) $c_1(\hat{L}_3) = -d \sum_{\nu=1}^{g} \frac{d}{d_\nu} da_\nu \wedge da_{g+\nu}$.

Proof. The push-forward map $\phi_{L_*} : H^2(X, \mathbb{Z}) \to H^2(\hat{X}, \mathbb{Z})$ is defined by the following commutative diagram

$$\begin{array}{ccc} H^2(X, \mathbb{Z}) & \xrightarrow{\phi_{L_*}} & H^2(\hat{X}, \mathbb{Z}) \\ {\scriptstyle P^{-1}}\downarrow & & \uparrow{\scriptstyle P} \\ H_{2g-2}(X, \mathbb{Z}) & \xrightarrow{\phi_{H_*}} & H_{2g-2}(\hat{X}, \mathbb{Z}) \end{array}$$

where P denotes the Poincaré duality of Section 1 (e) and ϕ_{H_*} in the lower row is the usual push-forward map in homology. According to equation (5) we have

$$\phi_{H_*}(\lambda_i) = H(\lambda_i, \cdot) = \begin{cases} d_i \varepsilon_{g+i} & \\ & \text{if} \\ -d_{i-g}\varepsilon_{i-g} & \end{cases} \begin{array}{l} i \leqq g \\ \\ i > g. \end{array}$$

Now let $\{D\}$ denote the homology class of D in $H_{2g-2}(X, \mathbb{Z})$. Then $P(\{D\}) = c_1(L)$ and thus, with equation (11)

$$\begin{aligned} \{D\} &= P^{-1}\left(-\sum_{\nu=1}^{g} d_\nu dx_\nu \wedge dx_{g+\nu}\right) \\ &= (-1)^{g-1} \sum_{\nu=1}^{g} d_\nu \lambda_1 \star \lambda_{g+1} \star \ldots \star \check{\lambda}_\nu \star \check{\lambda}_{g+\nu} \star \ldots \star \lambda_g \star \lambda_{2g}. \end{aligned}$$

Using (5) this implies

$$\begin{aligned} &\phi_{L_*}(c_1(L)) = \\ &= P\phi_{H_*}((-1)^{g-1} \sum_{\nu=1}^{g} d_\nu \lambda_1 \star \lambda_{g+1} \star \ldots \star \check{\lambda}_\nu \star \check{\lambda}_{g+\nu} \star \ldots \star \lambda_g \star \lambda_{2g}) \\ &= (-1)^{g-1} P \sum_{\nu=1}^{g} d_\nu \phi_{H_*}(\lambda_1) \star \phi_{H_*}(\lambda_{g+1}) \star \ldots \star \check{\lambda}_\nu \star \check{\lambda}_{g+\nu} \star \ldots \star \phi_{H_*}(\lambda_g) \star \phi_{H_*}(\lambda_{2g}) \\ &= (-1)^{g-1} P \sum_{\nu=1}^{g} d_\nu (-d_1^2 \varepsilon_{g+1} \star \varepsilon_1) \star \ldots \star (-d_\nu^2 \check{\varepsilon}_{g+\nu} \star \check{\varepsilon}_\nu) \star \ldots \star (-d_g^2 \varepsilon_{2g} \star \varepsilon_g) \\ &= (-1)^{g-1} d \sum_{\nu=1}^{g} \frac{d}{d_\nu} P(\varepsilon_1 \star \varepsilon_{g+1} \star \ldots \star \check{\varepsilon}_\nu \star \check{\varepsilon}_{g+\nu} \star \ldots \star \varepsilon_g * \varepsilon_{2g}) \\ &= -d \sum_{\nu=1}^{g} \frac{d}{d_\nu} da_\nu \wedge da_{g+\nu}. \end{aligned}$$

For the last equation we use Proposition 2.7 below and equation (11) applied to the dual abelian variety $\hat{X}$. Hence according to Proposition 2.3 there is an ample line bundle $\hat{L}_3$ on $\hat{X}$ (namely $\hat{L}_1^d$) such that $\phi_{L_*}(c_1(L)) = c_1(\hat{L}_3)$. This completes the proof of the proposition. $\square$

(d) The polarization $\hat{H}_4$. Let $\mathscr{P}$ denote the Poincaré bundle on $X \times \hat{X}$ that is the uniquely determined line bundle on $X \times \hat{X}$ such that (i) $\mathscr{P}|X \times \{\hat{x}\}$ is the line bundle on X represented by the point $\hat{x}$ and (ii) $\mathscr{P}|\{0\} \times \hat{X} = \mathscr{O}_{\hat{X}}$. According to Mukai [4] the Fourier transform of the ample line bundle L on X is defined to be the sheaf $\mathscr{F}(L) := p_{2_*}(\mathscr{P} \otimes p_1^* L)$ on $\hat{X}$. Here p_1 and p_2 denote projection maps of $X \times \hat{X}$ onto X and $\hat{X}$. It follows immediately from the base change theorem that $\mathscr{F}(L)$ is a vector bundle of rank $d = d_1 \cdots d_g$ on $\hat{X}$.

Proposition 2.6. (i) *The line bundle* $\hat{L}_4 := \det(\mathscr{F}(L))^{-1}$ *defines a polarization* $\hat{H}_4$ *on* $\hat{X}$.
(ii) $\phi_L^* \hat{H}_4 = dH$.

Proof. Mukai showed in [4] Proposition 3.11(1): $\phi_L^* \mathscr{F}(L) \simeq (L^{-1})^{\oplus d}$.
This implies (ii) since

$$\begin{aligned} \phi_L^*(\det(\mathscr{F}(L))^{-1}) &\simeq (\det \phi_L^*(\mathscr{F}(L))^{-1} \\ &\simeq (\det(L^{-1})^{\oplus d})^{-1} \\ &\simeq L^{\otimes d}. \end{aligned}$$

Moreover this implies (i), since the line bundle $\det(\mathscr{F}(L))^{-1}$ on $\hat{X}$ is ample if and only if its inverse image under the isogeny ϕ_L is ample. □

(e) Proof of the theorem. Propositions 2.3 and 2.4 imply $\hat{H}_2 = (d-1)!\hat{H}_1$. Propositions 2.3 and 2.5 imply $\hat{H}_3 = d\hat{H}_1$. Finally $\hat{H}_4 = \hat{H}_1$ follows from Propositions 2.2 and 2.6. □

(f) The type of the dual polarization. Let $H = c_1(L)$ be a polarization of type $(d_1, \ldots, d_g)$ on the abelian variety X and $\hat{H} := \hat{H}_1 = \hat{H}_4$ the dual polarization on $\hat{X}$. As usual let $d = d_1 \cdots d_g$ denote the degree of the polarization.

Proposition 2.7. *The dual polarization* $\hat{H}$ *is of type* $\left(\frac{d}{d_g}, \ldots, \frac{d}{d_1}\right)$ *on* $\hat{X}$ *and* $\varepsilon_{2g}, \ldots, \varepsilon_{g+1}, \varepsilon_g, \ldots, \varepsilon_1$ *is a symplectic basis of* $\hat{\Lambda}$ *for* $\hat{H}$.

In particular $\hat{H}$ is a principal polarization in the case of an elliptic curve X and $\hat{H}$ is of the same type as H if X is an abelian surface.

Proof. In the proof of Proposition 2.1 we showed for all $1 \leqq \nu,\ \mu \leqq g : \hat{H}(\varepsilon_\nu, \varepsilon_\mu) = \frac{d}{d_\nu d_\mu} H(\lambda_{g+\nu}, \lambda_{g+\mu})$. The same proof yields also $\hat{H}(\varepsilon_{g+\nu}, \varepsilon_\mu) = -\frac{d}{d_\nu d_\mu} H(\lambda_{g+\mu}, \lambda_\nu)$. Hence

$$\operatorname{Im} \hat{H}(\varepsilon_\nu, \varepsilon_\mu) = 0 \quad \text{and} \quad \operatorname{Im} \hat{H}(\varepsilon_{g+\nu}, \varepsilon_\mu) = \frac{d}{d_\nu} \delta_{\nu\mu}$$

for all $1 \leqq \nu, \mu \leqq g$. This implies the assertion. □

Remark 2.8. One could also define the dual polarization in such a way that its degree is small: Let H be a polarization of type $(d_1, \ldots, d_g)$ on X. In [2] we defined a dual polarization $\hat{H}_5$ on $\hat{X}$ by the property $\phi_L^* \hat{H}_5 = d_1 d_g H$. The polarization $\hat{L}_5$ is of type $\left(d_1, \frac{d_1 d_g}{d_{g-1}}, \ldots, \frac{d_1 d_g}{d_2}, d_g\right)$ (see [2], Theorem 2.4.1). Obviously $\hat{H}_1$ and $\hat{H}_5$ are related by $\hat{H}_1 = d_2 \cdots d_{g-1} \hat{H}_5$.

Remark 2.9. The theorem and its proof are more generally valid for complex tori with a polarization of some index. Recall from [2] that a polarization of index k on a complex torus X is by definition the first Chern class of a line bundle on X whose hermitian form is nondegenerate with exactly k negative eigenvalues. With this terminology an abelian variety is a complex torus admitting a polarization of index 0. Finally note that the dual polarization $\hat{H}$ has the same index as H.

References

[1] A. Beauville, Quelques remarques sur la transformation de Fourier dans l'anneau de Chow d'une variété abélienne. LNM **1016,** 238–260. Berlin-Heidelberg-New York 1983.

[2] C. Birkenhake and H. Lange, Complex Tori. Progr. Math. **177** (1999).

[3] H. Lange and C. Birkenhake, Complex Abelian Varieties. Grundlehren Math. Wiss. **302**. Berlin-New York 1992.

[4] S. Mukai, Duality between $D(X)$ and $D(\hat{X})$ with its application to Picard sheaves. Nagoya Math. J. **1,** 153–175 (1981).

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